SC1001: Introduction to Computational Thinking

Week 3: Boolean Logic and Branching

Jordan Boyd-Graber

Nanyang Technological University

Slides

Slides and updates will be posted at

`boydgraber.org/teaching/SC_1001/`

Overview

1. Predict the type and value of boolean expressions
2. Trace a program and predict what it prints
3. Evaluate an access-control condition step by step
4. Write a boolean rule for a smart home alarm
5. Trace named conditions in an access-control example
6. Follow an `if`/`elif`/`else` chain over a list
7. Optional: draw the control flow for an ATM menu

Discussion 1: Predict the Type and Value

For each expression, determine both its Python data type and its value.

Expression	Type	Value
`1 / 2`		
`1 // 2`		
`1 > 2`		
`True`		
`"True"`		
`3 < 3 and 3 / 0 > 0`		
`3 / 0 > 3 and 3 < 0`		
`3 > 0 or 3 / 0 != 1`		
`5 < 6 > 3`		

Discussion 2: What Prints?

```
a, b, c = 9, 10, 11
print(5 > a or b < 8 or 20 > c > 9)   # L1

a, b, c = 4, -1, 11
c, b, a = a // 2, int(c / 3), b + c
print(a, b, c)                        # L2

a = 8 + 9
if a > 8:
    print("bigger")                   # L3
    print("small")                    # L4
if a < 9:
    print("bigger")                   # L5
print("small")                        # L6
```

Output

Predict before running.

Discussion 3: What Is the Output?

```python
is_admin = True
is_logged_in = True
has_permission = False

access_granted = is_logged_in and (is_admin or has_permission)

if access_granted:
    print("Access granted.")
else:
    print("Access denied.")
```

Discussion 4: Write the Boolean Rule

Suppose we use these variables:

- `motion`: motion is detected
- `armed`: the alarm system is armed
- `door_open`: a door is open
- `window_open`: a window is open

Write a boolean expression that triggers the alarm exactly when there's motion, the system is armed, and either the window is open or the door is open.

Discussion 5: What Is the Output?

```python
user_role = "administrator"
request_content = "delete record"

is_admin = (user_role.startswith("admin"))
is_manager = (user_role == "manager")
contains_delete = ("delete" in request_content)
contains_record = ("record" in request_content)

access_granted = (is_admin or is_manager) and contains_delete and contains_record

if access_granted:
    print("Access granted.")
else:
    print("Access denied.")
```

Discussion 6: What Is the Output?

```python
items = ["apple", "carrot", "durian", "tomato",
         "banana", "spinach", "orange", "broccoli"]
known_fruits = ["apple", "banana", "orange", "tomato"]
known_vegetables = ["carrot", "spinach", "broccoli", "tomato"]

num_fruit = num_veg = num_unknown = 0

for item in items:
    if item in known_fruits:
        num_fruit = num_fruit + 1
    elif item in known_vegetables:
        num_veg = num_veg + 1
    else:
        num_unknown = num_unknown + 1

print("Fruits:", num_fruit)
print("Vegetables:", num_veg)
print("Unknown:", num_unknown)
```

Discussion 7 (Optional): ATM Menu Skeleton

```
balance = 1000.0

while True:
    print("""\nATM Menu:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit""")

    choice = int(input("Enter your choice (1-4): "))
```

How should we branch on `choice`?